

End Semester Examination
December 2024

Name of the Program: B Tech Semester: 1st
Name of the Course: Engineering Chemistry
Course Code: TCH 101

Time: 3 Hr

Maximum Marks: 100

Note:-

- (i) All questions are compulsory.
- (ii) Answer any two sub questions among a, b & c in each main question
- (iii) Total marks in each main question are twenty.
- (iv) Each question carries 10 marks

Q1 (20 marks)		CO 1
a)	Explain the magnetic nature of HF and NO molecule with the help of molecular orbital diagram and also find out the bond order.	
b)	What is the necessary condition for a molecule to be IR active? Give the difference between UV- Visible and IR Spectroscopy.	
c)	Define metallic bonding in metals. Explain the band theory by taking the example of Sodium metal. How many vacant orbitals are present in Sodium metal?	
Q2 (20 marks)		CO 2
a)	I. Explain Zeolite method of water softening and its regeneration. Write its advantages and limitations. II. Total Hardness of 20,000 liters of water was completely removed by zeolite softener. It required 40 Liters of NaCl solution containing 10.5% NaCl for regeneration. Calculate the hardness of water sample.	
b)	Differentiate: i. Cold and Hot Lime Soda Process ii. Scales and Sludges	
c)	A water sample contains the following impurities per litre. Ca(HCO ₃) ₂ = 40.5mg/l, Mg(HCO ₃) ₂ = 36.5mg/l, CaSO ₄ =34.05mg/l, MgSO ₄ =30.05mg/l, CaCl ₂ =27.75mg/l, NaCl=35.0mg/l Calculate temporary and permanent hardness in water and Quantity of lime and soda required in kg for softening 10,000 liters of water.	
Q3 (20 marks)		CO 3
a)	Give structure, properties and applications of I. PVA II. Kevlar III. Nylon 6,6	
b)	What is conducting polymer? Classify conducting polymers by giving suitable examples and write its applications.	
c)	Describe the classification of polymers on the basis of mode of formation. Explain the free radical mechanism of formation of polyethene from ethene.	
Q4 (20 marks)		
a)	Describe the instrument by which the calorific value of a solid liquid non volatile liquid is determined. Explain all the corrections and derived the formula for calorific value determination.	

b)	Define calorific value of a fuel and determine calorific value of a fuel (GCV and LCV) (assuming latent heat of steam=600cal/g) when the following results were recorded by a bomb calorimeter: Weight of Fuel Burnt = 1.50 gm. Water Equivalent of Calorimeter = 600 gm. Weight of water taken in calorimeter = 2500 gm. Initial Temperature of water =25.15° C. Final Temperature of water = 28.98°C. Acid Correction=10 cal. Fuse wire Correction = 5 cal. Cotton tread correction = 6 cal. Cooling Correction=0.25°C. The fuel contains 6 % Hydrogen.	CO4
c)	Write a short note on the following: (i) Differentiate between LPG and CNG (ii) Compare and contrast between solid, liquid and gaseous fuels	
Q5 (20 marks)		
a)	What is electrode potential? Write note on electrochemical series and its applications.	
b)	Explain the following: I. Rusting of iron is quicker in saline water than in ordinary water. II. Can we use a copper vessel to store 1 M AgNO ₃ solution? Given that: $E^{\circ}_{\text{Cu}^{2+}/\text{Cu}} = + 0.34 \text{ volt}$ and $E^{\circ}_{\text{Ag}^{+}/\text{Ag}} = 0.80 \text{ volt}$ III. Why do we use inert electrolytes like KCl, KNO ₃ , and NH ₄ Cl in a salt bridge? IV. Copper does not react with dil HCl whereas Zinc immediately produces hydrogen gas.	CO5
c)	For the cell; $\text{Mg(s)} \text{Mg}^{2+}(2 \text{ M}) \text{Cu}^{2+}(0.2\text{M}) \text{Cu(s)}$ [$E^{\circ}_{\text{Mg}^{2+}/\text{Mg}} = -2.37 \text{ V}$ and $E^{\circ}_{\text{Cu}^{2+}/\text{Cu}} = 0.34$] (i) Identify the cathode and the anode. (ii) Write the oxidation and reduction reactions. (iii) Calculate the EMF of cell.	